HW12

1. $w = \frac{w_s}{2} = \frac{10000\pi}{2} = 5000\pi$

2. $T = \frac{2\pi}{2W} = \frac{2\pi}{2\times 1000\pi} = 10^{-3}$

$0.5\times 10^{-3} < 10^{-3}$ $\quad 10^{-4} < 10^{-3}$

$\therefore$ a, c guarantee

3. (a) $w_N = 2\times 4000\pi = 8000\pi$

(b) $w_N = 2\times 4000\pi = 8000\pi$

(c) $w_N = 2\times 4000\pi = 8000\pi$

$w_N = 2\times 8000\pi = 16000\pi$

4. (a) $w_N = w_0$

(b) $\frac{dx(t)}{dt} \Rightarrow jw X(jw)$

$w_N = w_0$

(c) $x^2(t) \Rightarrow \frac{1}{2} x(jw) * x(jw)$

$w_M = 2w_{M0}$

$\therefore w_N = 2w_0$

(d) $[-\frac{w_0}{2}, \frac{w_0}{2}] \to [-\frac{3w_0}{2}, -\frac{w_0}{2}] \cup [\frac{w_0}{2}, \frac{3w_0}{2}]$

$w_N = 2\times \frac{3w_0}{2} = 3w_0$

6. $w' = w_1 + w_2$

$T = \frac{2\pi}{2w'} = \frac{\pi}{w_1+w_2}$

7. $x_p(t) = \sum x(nT_s)\delta(t-nT_s)$

$x_0(t) = x_p(t) * h_0(t)$ $\quad x_1(t) = x_p(t) * h_1(t)$

$h_0(t) = u(t) - u(t-T_s)$

$H_0(jw) = \int_0^{T_s} e^{-jwt} = e^{-jw\frac{T_s}{2}} \frac{2\sin(\frac{wT_s}{2})}{w}$

$h_1(t) = \begin{cases} 1-|t|T_s, & |t|<T_s \\ 0, & |t|>T_s \end{cases}$

$H_1(jw) = \frac{1}{T_s}\left(\frac{2\sin(\frac{wT_s}{2})}{w}\right)^2 = T_s\left(\frac{\sin(\frac{wT_s}{2})}{\frac{wT_s}{2}}\right)^2$

$X_1(jw) = H_f(jw) X_0(jw)$

$H_f(jw) = \frac{H_1(jw)}{H_0(jw)} = e^{jw\frac{T_s}{2}} \frac{\sin(wT_s/2)}{wT_s/2}$

8. (a) $w = 5\pi$ $\quad T_s = 0.2$

$2W = 10\pi$

$\sin(5\pi n T_s) = \sin(\frac{5\pi n}{5}) = \sin(n\pi) = 0$

Yes, aliasing occurs, $k=5$ lost

(b) $\frac{\pi}{T_s} = 5\pi$ $\quad w_0 = \pi$ $\quad T_0 = 2$

$(\frac{1}{2})^k \sin(k\pi t) = (\frac{1}{2})^k \frac{e^{jk\pi t} - e^{-jk\pi t}}{2j}$

$a_k = \frac{(\frac{1}{2})^k}{2j}$ $\quad a_{-k} = -\frac{(\frac{1}{2})^k}{2j}$

$w_s = 10\pi = 10w_0$

$g(t) = \sum_{k=1}^{4} (\frac{1}{2})^k \sin k\pi t$

$g(t) = \frac{1}{2}\sin\pi t + \frac{1}{4}\sin 2\pi t + \frac{1}{8}\sin 3\pi t + \frac{1}{16}\sin 4\pi t$

9. $Y(jw) = \begin{cases} 1 & |w|<50\pi \\ 0 & |w|>50\pi \end{cases}$

$X(jw) = \frac{1}{2\pi} Y(jw) * Y(jw)$

$|w| < 100\pi$

$T_s = \frac{2\pi}{w} = \frac{2\pi}{150\pi} = \frac{1}{75}$

$G(jw) = \frac{1}{T_s}\sum X(j(w-mw_s)) = 75X(j(w-m150\pi))$

$G(jw) = 75X(jw)$

$\therefore w_0^{max} = 50\pi$

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